Dynamic Player Clustering for Replacements, Synergy, and Compatibility

Team Members:

20BDS0392 (Pranjal Timsina)

20BDS0067 (Atharva Shekatkar)

20BCE2850 (Nida Ghani)

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Professor: Dr. P. Mohankumar

Abstract: *In the years from 1863 - when association football first started, football's growth has been nothing short of miraculous. The game which was once won based on physical strength, speed and sharpness, has now transformed to something highly tactical, positional and data driven. FIFA, a simulation of this sport which gets revised each year in September, models players and teams based on their real world performances; the high quality data they collect annually, and update it on January/February, can be used for scouting, replacements, positional changes and tactical experimentation. In this work, we aim to explore how replacements for players can be identified and what alternate positions a player can play in without having to change their playstyle a lot. Furthermore, we can identify the key positions for which a given team does not have any player. Some methods proposed for clustering players are: projective adaptive resonance theory algorithm for cluster analysis, a neural network based algorithm efficient for clustering high-dimensional data, k-means clustering, expectation minimization and hierarchical clustering methods. Principal Component Analysis is also proposed to identify and visualize correlations between the position of the players. In this paper, we have proposed a new model that has dynamic cluster size and a dynamic number of clusters that suits the given problem at hand.*

Keywords: *Sports, Football, Positional Play, Clustering, KMeans, Hierarchical Clustering, Tree, Divisive Clustering, Agglomerative Clustering*

1. Introduction

Positional play in football is nothing short of interesting - if you zoom out you'll find two kinds of players: goalkeepers and outfield players - this resolution is by no means adequate for appropriate data analysis, thereby making us look a little more closely leading us to discovering attackers, defenders, midfielders and goalkeepers, which still is not detailed enough. Upon closer inspection, each position discovered in the previous iteration can and will be subdivided. If we continue this an arbitrarily high number of times, we will have a different class for each and every player in the game, which does us no favour. So, it is important to decide how general or specific the positions are going to be. Great teams that go on to dominate the leagues for decades have a balance of generalisation and specialisation such that the team as a whole does not depend on the exact playstyle of one single player, while also not being so generalised that any player can fill the void left by his predecessor - take for example FC Barcelona, who once used to be dominant for huge stretches of time, but in the past decade, they had become too reliant on Lionel Messi, who is impossible to replace, leading the team to fall off a cliff; on the contrary, Manchester United managed by Sir Alex Ferguson from 1986 to 2013 were dominant throughout the period for they didn't rely on the specific attributes of a single player, but were not so simply organised that anyone could easily understand their playstyle. [3]

The aim of this introduction is to highlight the importance for a team to be able to find replacements for each player, so that the team does not struggle in the absence of an individual. Upon clustering the players, we can identify outliers, which can add a new element to a team but are hard to replace, and the players which can easily be replaced. The inference from the clustering can be used in the following way: the core of the team should comprise of players who can be replaced rather easily, each team should have some trump cards which can add different elements to their game, who however should not be relied on too much.[6]

2. Literature survey

Positional Play in Football explores how players cover space, or the lack thereof [5]. If one looks at this domain with too keen an eye, they could argue that there are infinite permutations of positional play, which I agree with; therefore, for the sake of brevity and the concise nature of this study, a handful of roles will be thoroughly examined.

Positional play simplifies the game based on the positions, spaces, objective and time. A very crude way to put this would be that any action a particular player takes is a function of his position, the opposition's position, the space of the field the position is in, the objective he has, and the time of the game. For example, let us consider a center back (the player who plays in the center of the defence) - under the given circumstance - there is a corner kick, all the opposition players are defending their goal post, there is empty space in the far-post, the defender's team desperately need to score a goal, and it is the final minutes of the match - naturally, the defender should occupy the far post of the goal and try to score a goal.[17]

How positional play fits into our study is that a certain mould of player can carry out the action he is supposed to take as we established in the "function" we defined above; for instance, if the centre back in the situation described above was not strong in the air, and could not head the ball properly, he would fail to fulfil his duty. Thereby finding the right players for a particular role is crucial [19]. Trying to slot in a wrong player at a given position has time and again proven to be disastrous - yes, there have been a handful instances where the decision was pure genius, but this is too risky in a trillion dollar industry like football; Philippe Coutinho, a player transferred to Barcelona for a sum of 160 million Euros, was forced out wide, where he couldn't perform as well as he used to playing in a more central role - which let both the player and the team down on occasions more times than a club of that calibre should be comfortable to admit [20]. Let us consider a team, Manchester City - they rely heavily on their attacking midfielder and playmaker Kevin De Bruyne; consider the fact that the player they rely on so heavily might decide to leave, or might get injured at any moment - since the consequences would be disastrous, a replacement must be lined up at all times.
Finding replacements the orthodox way - sending scouts over the different countries - is not very scalable. Here is where the use of data can result in an as good or better set of possible replacements compared to what the scouts prepare.
Existing literature [7] has proposed multiple clustering methods including K-means and Gaussian Clustering[2]. The research paper, Soto-Valero, César. (2017). A Gaussian mixture clustering model for characterising football players using the EA Sports' FIFA video game system consists of a model-based clustering method for selecting professional football players using multivariate data of performance[9]. Here, they used principal component analysis to reduce data dimensions and then Gaussian finite mixture clustering to group the data. Their main aim was to select players according to their clustered characteristics and then label them according to different roles. Also their purpose was to discover the common characteristics of players related to each role, as well as the profiles of border line players.

However due to misclassification during the clustering process, some defensive and offensive players still had relatively high average scores in activities associated with goalkeepers, such as diving or handling and raise doubts on the reliability of these scores.

This work used Gaussian finite mixtures in order to group and classify players according to performance attributes using probabilistic and statistical criteria and supported the conception that each position of players in football is defined by specific performance indicators.

We believe that flat K-means clustering is too simple to capture every aspects of player positions, while gaussian clustering's fundamental problem is that it assumes that the skill of players is normally distributed - which by domain knowledge we disagree with; yes, it can be argued that the crude physical factors like height, weight, strength, and speed might be normally distributed, but the the other metrics that help judging players playing in a certain position are usually heavily skewed in top-flight football. On top of that, the sample size of top quality players that top clubs are looking for is generally small, and it is a basic fact that normal distribution normally reveals itself when the sample size is large. [1]
Upon examining other clustering methods [13], modified divisive clustering [14], which is described below under *Proposed Model*, is seemingly the best fit for what we aim to do in this project.

3. Objective of the project

Time has shown us over again and again that finding replacements for players is not an easy task. A data based approach to finding the replacements of players is a much better way than mere speculation. In this project, we aim to use dynamic clustering to find the best replacements based on data.

4. Innovation in the project

Despite the conceptual simplicity and efficiency of flat clustering, it has a number of drawbacks. While older studies have effectively used K-means clustering or Guassian clustering [12] in clustering players, we believe that the results are severely limited, because positional similarities are not binary (similar or dissimilar), but instead they have a degree of similarity or dissimilarity which is better encapsulated by a hierarchy. Another drawback of flat K-means clustering is that we must specify the number of clusters beforehand. There are two possible solutions that we have proposed - modified agglomerative clustering and modified divisive clustering, which we have ***modified to have a dynamic cluster size at runtime***.[16]

Modified agglomerative clustering works on the following principle. We first choose *n*, the size of the smallest cluster in the hierarchy. All the nodes in the tree (hierarchy) can possibly have n^i or i^n *records.* As new clusters are generated from combining smaller clusters, the new one is added to a Cluster Tree. The technicalities of this data structure can be rather detailed, but this is a sufficient abstraction for modified agglomerative clustering. [9]

Modified divisive clustering[11] in spirit is quite similar to modified agglomerative clustering, but there are significant differences that will have to be gone over. We start with the entire dataset as a single cluster, then repetitively divide the dataset into sub clusters, whilst forming a tree-like structure of depth *n*, where each cluster has a minimum of *i* records.

Each of the two methods above have their own set of drawbacks and benefits. While the deterministic nature of modified agglomerative clustering, unlike the divisive method, is lucrative, the time complexity of $O(n^3)$ and the space complexity of $O(n^2)$ is something to take into consideration. [15]

Taking into consideration the pros and cons of the methods described above, we plan to use modified divisive clustering where a root node consisting of all the records in the data set is then recursively partitioned until a maximum of n^i clusters are formed where each cluster has at least m records. [8]

While this sounds awfully similar to traditional hierarchical clustering, it couldn't be further from the truth. A big drawback is that after deploying the model, we cannot change the cluster size - it is fixed when the algorithm completes running.[4]

Our proposed model addresses this very issue. The model deployed by our method encapsulates a hierarchy which we can zoom in and zoom out on depending on how much we want the cluster size to be.

A valid question to be asked is why use recursive divisive clustering that outputs a tree like structure - won't it just increase the computation power required to cluster n players, or what added benefit does it have compared to K-means or agglomerative clustering?

K-means clustering requires us to specify n, the number of clusters we want, while on the other hand agglomerative clustering does not - modified divisive clustering falls somewhere in the spectrum of us having to specify the number of clusters, or not. We can specify a number d, and the number of clusters formed will be at most $i^d - 1$. Depending on how much we wish to zoom in or out (given $i = 2$), we can have anywhere from 1, 2, 4, 8, … , $i^d - 1$ clusters. For instance, let us consider that we have used K-means clustering and have specified *n_clusters* to 100, and later when checking the players in the same cluster as, say, Jack Grealish, we get 5 other players. What if we want more players similar to him? Here, either we have to re-run the algorithm with *n_clusters* < 100, and hope that the cluster size will be larger, or we can use our proposed algorithm - modified divisive clustering - that completely eliminates this problem.

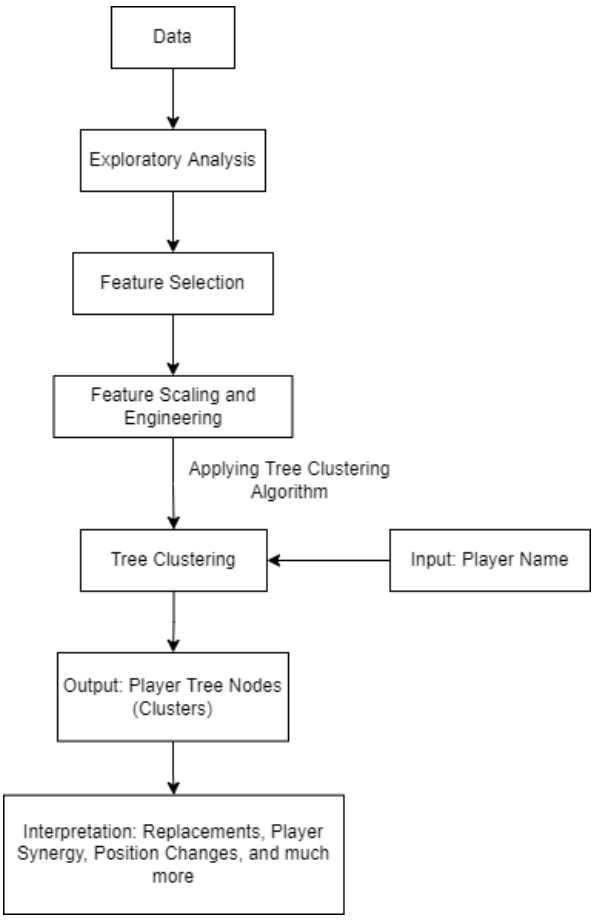


Fig. 1. Flow of data in the proposed model

The figure above illustrates the flow of data in the proposed model. The figure is obtained after a couple layers of abstraction, but is descriptive enough to show how the proposed model works. After a cluster tree is generated, the player name is given as input and all the nodes and subnodes (clusters and sub-clusters) are shown. The interpretation of the results is only limited by the user's understanding of the domain and their creativity.

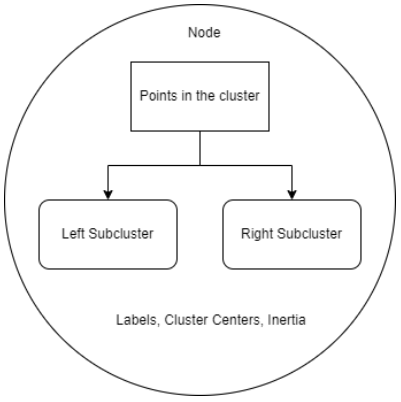


Fig. 3. Structure of a Node in the Tree

In the above diagram, we have presented an abstraction of a general node. A typical node in our tree is an abstraction over a cluster, its children and the attributes of the cluster. The children (left subcluster and right subcluster in the diagram) are nodes themselves, and follow a similar structure. The leaf nodes have a NULL pointer to their children. There are three attributes: labels, cluster centers and inertia stored along with the cluster points and the children. Labels specify whether a given point in the cluster belongs to the left subcluster or the right subcluster. The cluster centers are the cluster centers of the two sub clusters. Inertia is the sum of squared distances of samples to their closest cluster center.

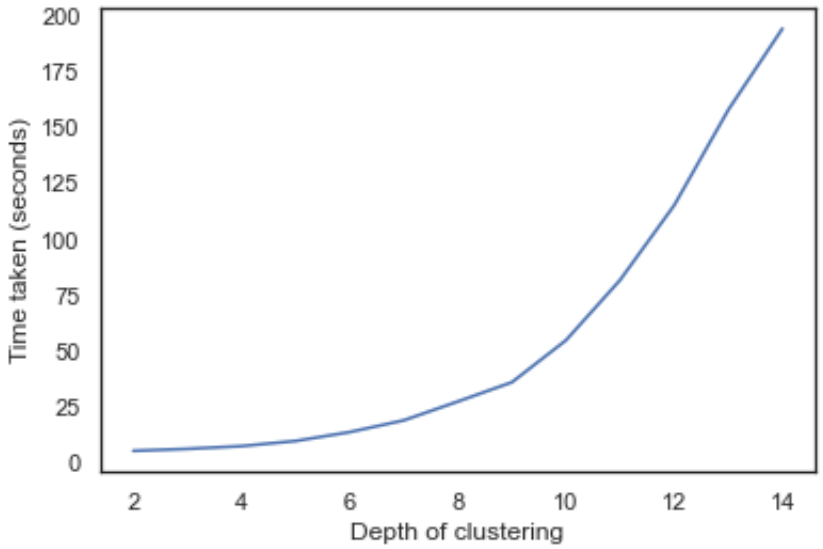


Fig. 3. Time taken to cluster upto depth i

It is evident in the graph given above that the runtime does not scale well with the depth of the cluster tree. This drawback is rather insignificant as once the cluster tree is formed, re-clustering to form clusters of a larger or smaller size won't have to be done because of the nature of the cluster tree we provide.

At first glance, the proposed model sounds quite similar to hierarchical clustering, however, it is quite different from it. Upon finishing the clustering, hierarchical clustering, regardless of whether it is divisive or agglomerative, produces n different clusters of size a to b, where the smallest cluster is of size a and the largest is of size b. When using the trained "model" after deployment, the clusters are fixed - you cannot increase the cluster size, or decrease it, neither can you increase or decrease the number of clusters. The model that we propose completely overcomes this issue. The cluster size and the number of clusters is dynamic - the user can specify if he wants a few large clusters, or many small clusters or anything in between.

5. Work done and implementation

a. Methodology adapted

The dataset used in this project was rather big, so it was critical to understand the dataset first. Exploratory data analysis was done, and conclusions were derived from the dataset. The findings from the conclusions are explained in section seven.

The hardware requirements for the project are a mid tier CPU and GPU. Improving the CPU or GPU used would only marginally affect the time taken. The software requirements are quite basic - Google Colaboratory was used with a python kernel. The libraries used were numpy, matplotlib, seaborn, pandas, and sklearn.

b. Dataset used

An open source dataset provided on Kaggle titled FIFA 21 players was used. The dataset was based on FIFA 22, a football simulation created by EA, a company based in Northern California. This project is not based on any reference project.

c. Screenshot and demo

The relevant screenshots are given below:

```
class Node:
  def __init__(self):
    self.children = None
    self.data = None

  def __repr__(self):
    return f"Cluster Size: {len(self.data)}"

  def __str__(self):
    return f"Cluster Size: {len(self.data)}"
```

```
def cluster_wrapper(data_frame, player_name, min_rating, min_cluster_size):

  # If min rating < player.rating then min rating = player.rating
  if ((data_frame[data_frame['long_name'] == player_name]['overall'] < min_rating).all()):
    min_rating = int(data_frame[data_frame['long_name'] == player_name]['overall'])

  # Filter players using min rating
  # data_frame = data_frame[data_frame['overall'] > min_rating]
  # Seperate the labels and features

  labels = data_frame["sofifa_id"]
  features = data_frame.drop("sofifa_id", axis=1)

  root = cluster(features, labels, 2)
  labels['labels'] = root.labels_
  return data_frame, root

def cluster(X, y, n_clusters):
  kmeans = KMeans(n_clusters=n_clusters, random_state=0, algorithm="elkan").fit(X, y)
  return kmeans
d, k = cluster_wrapper(data_frame=predictors, player_name='Robert Lewandowski', min_rating=80, min_cluster_size=10)
cluster_map = pd.DataFrame()
cluster_map['data_index'] = predictors.index.values
cluster_map['cluster'] = a.labels_
cluster_map['cluster'].value_counts()
```

```
def get_children(data, kmeans):
    data['labels'] = kmeans.labels_
    left_child = data[data['labels'] == 0].copy()
    right_child = data[data['labels'] == 1].copy()
    return left_child, right_child
```

```
cluster = None
def has_player(cluster, player):
    try:
        player_id = name_id_dict[player]
        # print(player_id)
        return cluster.data[cluster.data['sofifa_id'] == player_id].any().any()
    except:
        return False
```

```
print_player_clusters("Harry Maguire", root)
```

```
Cluster 0: Size: 19239
Cluster 1: Size: 17107
Cluster 2: Size: 9330
Cluster 3: Size: 4340
Cluster 4: Size: 681
Cluster 5: Size: 113
Cluster 6: Size: 84
Cluster 7: Size: 60
Cluster 8: Size: 33

Custer 9:
R. Varane
A. Laporte
M. Škriniar
S. de Vrij
M. de Ligt
Piqué
H. Maguire
L. Hernández
C. Lenglet
J. Gomez
D. Upamecano

Custer 10:
R. Varane
A. Laporte
S. de Vrij
Piqué
H. Maguire
C. Lenglet

Custer 11:
S. de Vrij
Piqué
H. Maguire
C. Lenglet
```

6. Interpretation and use of the results

The use cases for this algorithm can be as simple or as complex as we can imagine; a trivial way to use it would be to find out potential replacements. Also, players changing positions in their careers is quite common, sometimes position changes happen for the better and sometimes for the worse - this algorithm can help us figure out what positional changes make sense and what does not. For example in a cluster full of defenders, if there is a midfielder, then it is an indicator that the player can potentially change his position to a defender and possibly thrive there. A much more advanced way of using the results is finding out the complementary cluster - a cluster whose players, when playing with a player *X*, bring out the best out of *X*. An anecdote would be the noticeable improvement in how Jadon Sancho plays when playing with Donny Van de Beek. Here, upon examining data metrics like xG and xA of a given player when playing with another player, we can do cluster analysis to figure out which players should play together in a team and who should not. The same technique can also be used for scouting new players and checking beforehand whether a given player fits in the team well or not.

```
print_player_clusters("Harry Maguire", root)
```

Cluster 0: Size: 19239	Custer 9:	Custer 10:
Cluster 1: Size: 17107	R. Varane	R. Varane
Cluster 2: Size: 9330	A. Laporte	A. Laporte
Cluster 3: Size: 4340	M. Škriniar	S. de Vrij
Cluster 4: Size: 681	S. de Vrij	Piqué
Cluster 5: Size: 113	M. de Ligt	H. Maguire
Cluster 6: Size: 84	Piqué	C. Lenglet
Cluster 7: Size: 60	H. Maguire	
Cluster 8: Size: 33	L. Hernández	Custer 11:
	C. Lenglet	S. de Vrij
	J. Gomez	Piqué
	D. Upamecano	H. Maguire
		C. Lenglet

Fig. 4. The cluster tree nodes which contain Harry Maguire

Here cluster 0 is the root of the hierarchy. Cluster 0 has two children A and B. Only one of A and B contains the given player *Harry Maguire*. Cluster 1 is one of those children who fulfills this criteria. Abovc, the children, grandchildren clusters and so on which contain the given player are displayed. Moving from cluster 0 → 1 → 2 → … → 11 is like zooming in to get a smaller, more cohesive cluster, while moving from 11 → … → 1 is like zooming out to get a larger cluster size. Cluster 11 shows the players most similar to Harry Maguire while cluster 10 shows more number of players similar to him at the cost of a relatively lower similarity.

7. Experimental Analysis

The objective of this empirical analysis of the dataset is to find out the distributions of various features in the dataset.

First, let us have a look at the crude physical features of players in the dataset.[10]

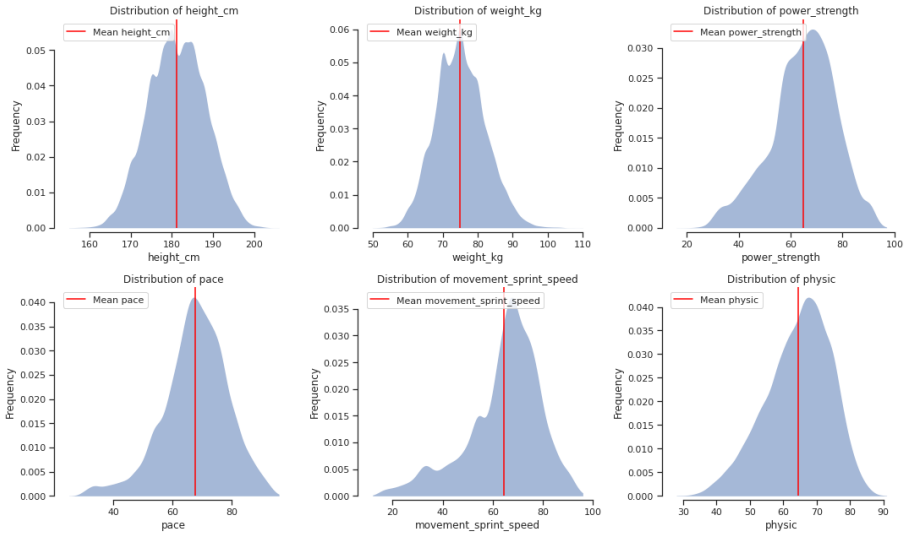


Fig. 5. Distribution of physical features of players in the dataset.

Here, height is the closest to a normal distribution, which isn't surprising at all. Players actively train to attain a specific weight, strength, pace and physique which brings skewness in the distribution.

A hypothesis that the negative skewness of pace, strength, physic and sprint speed can be attributed to the older players can be put forth.

We can look at the average pace, strength and other attributes in different age groups to test this hypothesis.

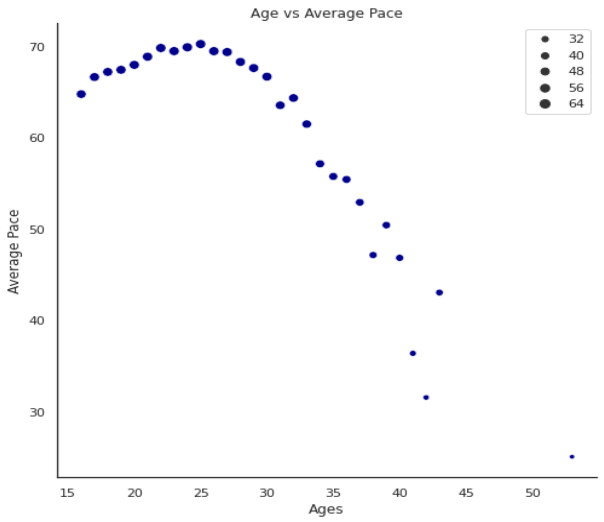


Fig. 6. Plot of Average pace of the players vs their age.

In the figure above (Average pace of the players vs their age), we can see a steady drop off after the age of 25 to 27. The other attributes, too, drop off in a similar fashion at the same age.

The key takeaway from this is that to replace a given player in a team, it wouldn’ t be very wise to invest in a player over the age of 27, unless it's for a short to medium term. Ideally, the younger the player the more time he can be at his physical peak while playing for a club.

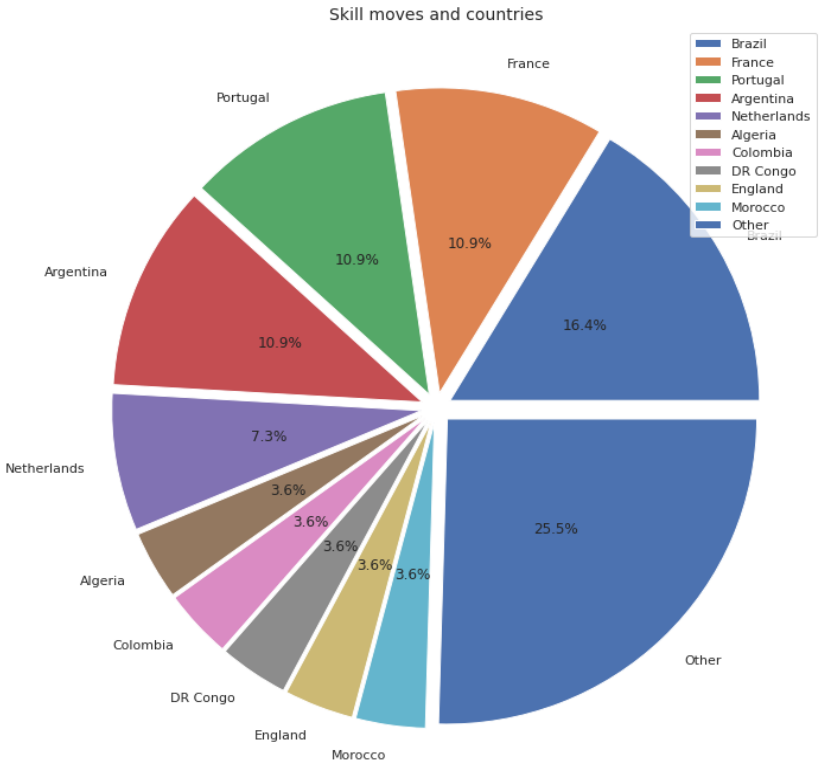


Fig. 7. Pie chart of players with 5 star skill moves grouped by country

To understand the figure above, we need some background information. In the dataset, the *skill moves* of a player is rated from 1 star to 5 stars, which represents the player's ability to skilfully dribble the ball, often in a flamboyant and effective way. Some famous players with 5 star skill moves are Cristiano Ronaldo, and Neymar who are known to showboat during the match. Having 4 or 5 star skill moves is a desirable attribute for midfielders and attackers, and even fullbacks on some occasions. On multiple occasions it has been proven that for any football team, 5 star skillers are very marketable for a team - this fact coupled with the efficacy of these players on the field, makes these players highly sought after. The pie chart above illustrates that a quarter of the players with 5 star skill moves are from Brazil - which is not surprising given Brazil's motto of *joga bonito,* which translates to "play beautifully". This attribute can be highly weighted when searching for attackers and midfielders in the algorithm we use.[9]

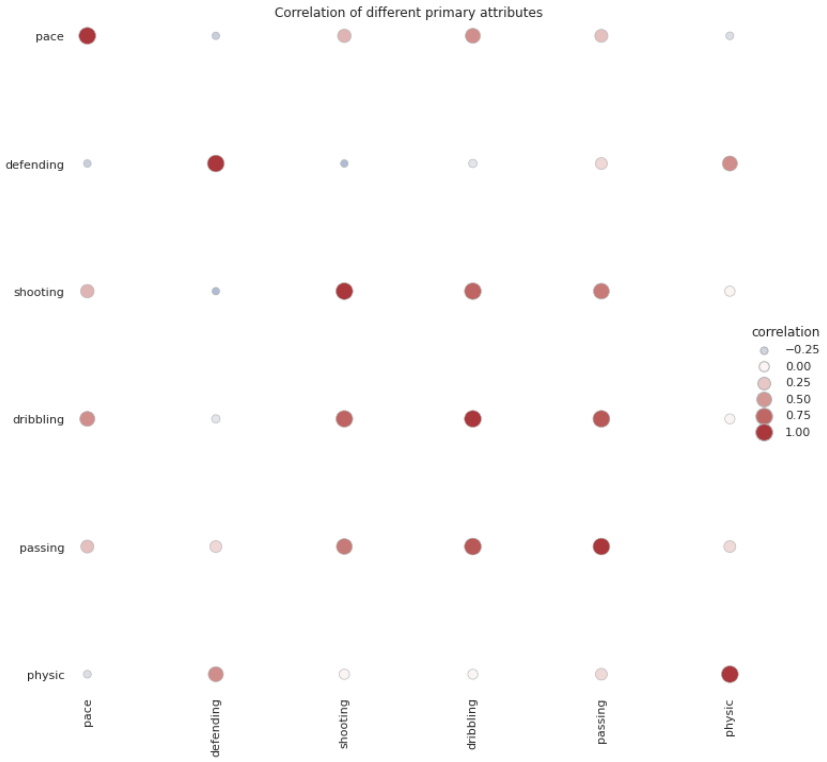


Fig. 8. Karl Pearson correlation matrix of the primary attributes of football players.

The figure above is a Karl Pearson correlation matrix of the primary attributes of football players: pace, dribbling, defending, shooting, passing, and their physique [18]. The figure illustrates the following strong correlations:

- Defending & Physique
- Shooting & Dribbling
- Shooting & Passing
- Dribbling & Passing

The correlations above are quite intuitive, too, players who are good at defending usually have a strong physique, while midfielders and attackers are fast, agile, and can shoot well.[9]

The performance of the model that we have proposed is significantly better than the existing models as presented in the literature given in the references, primarily because of one reason - we do not use a fixed number of clusters or fixed cluster sizes. When you fix either of the variables, the given problem at hand requires fine tuning of at least one of the two parameters stated above, which is hard, no, impossible to accomplish with a time complexity that makes sense. Our method which has a tree, a real hierarchy, available in runtime, which can zoom in and out of clusters (increase the cluster size or decrease it, and/or increase the number of clusters or decrease it), which encapsulates the fact that positions in football can be zoomed in or out on, as there are very specific roles, and very general roles at the same time.

8. References

1. Shibata, R. T.,Pedrini, H. Data Mining Approach to Prediction of Nominee Soccer Players for Ballon d'Or Award, 2018.
2. Soto-Valero, César. A Gaussian mixture clustering model for characterizing football players using the EA Sports' FIFA video game system, 2017.
3. Diquigiovanni, Jacopo, & Scarpa, Bruno. Analysis of association football playing styles: An innovative method to cluster networks, 2019.
4. Tosato, Marco, & Wu, Jianhong. An application of part to the football manager data for players clusters analyses to inform club team formation, 2018..
5. Wijngaard, Gijs. (n. d). Clustering soccer players: investigating unsupervised learning on player positions, 2020.
6. Nunes, Sérgio, & Sousa, Marco. (n. d). Applying Data Mining Techniques to Football Data from European Championships, 2018.
7. Manning, Christopher D. & Raghavan Prabhakar, Schutze, Henrich. Introduction to Information Retrieval, 2008.
8. Gao, Jing. (n.d). Clustering: Hierarchical Methods.
9. Arruda Moura, F.; Barreto Martins, L. E., & Augusto Cunha, S. Analysis of football game-related statistics using multivariate techniques. Journal of Sports Sciences, 2013.
10. Bloomfield, J.; Polman, R.; Butterly, R., & O'Donoghue, P. Analysis of age, stature, body mass, BMI and quality of elite soccer players from 4 European leagues. The Journal of Sports Medicine and Physical Fitness, 2005.
11. Baio, G, Blangiardo, M. Bayesian hierarchical model for the prediction of football results. Journal of Applied Statistics, 2010.
12. Kaufman, L, Rousseeuw, PJ. Finding Groups in Data: An Introduction to Cluster Analysis, volume 344. New Jersey: John Wiley & Sons, 1990.
13. Peter J Rousseeuw, Silhouettes: a graphical aid to the interpretation and validation of cluster analysis, Journal of computational and applied mathematics, vol. 20, 1986.
14. Cohen-Addad, V., Kanade, V., Mallmann-Trenn, F., & Mathieu, C.. Hierarchical Clustering. Journal of the ACM (JACM), 66, 1 - 42, 2019.
15. Cohen-Addad, V., Kanade, V., Mallmann-Trenn, F., & Mathieu, C. Hierarchical Clustering: Objective Functions and Algorithms. SODA, 2018.
16. Roy, S., & Sasmal, B. Integration of Hierarchical Clustering method and Dendrogram method with Expectation Maximization for identification of the best player cluster. 2021 7th International Conference on Optimization and Applications (ICOA), 1-8, 2021.
17. Wensveen, C.J. Classification of playing styles in football: The use of ball action data, 2016.
18. Di Salvo, V., Baron, R., Tschan, H., Calderón Montero, F.J., Bachl, N., & Pigozzi, F. Performance characteristics according to playing position in elite soccer. International journal of sports medicine, 28 3, 222-7 , 2007.
19. Martín-García, A., Casamichana, D., Díaz, A.G., Cos, F., & Gabbett, T.J. Positional Differences in the Most Demanding Passages of Play in Football Competition. Journal of sports science & medicine, 17 4, 563-570, 2018.
20. Fomini Tendonkeng, J., Benoît Nguelefack, T., Brice Assomo Ndemba, P., Richard Guessogo, W., Mekoulou Ndongo, J., & Temfemo, A. Physical and Physiological Characteristics of Cameroon Professional Soccer Players According to Their Competitive Level and Playing Position. International Journal of Sports Science, 6, 8-11, 2021.